

Aura Research Paper Writing Pipeline

Current state assessment · Architecture proposals · Implementation roadmap

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1. What Exists Today

1.1 :research <topic> — Single-Agent Pass

File: `src/research/research.ts` (110 lines). Inspired by ByteDance's **DeerFlow** SuperAgent harness.

- One agent, 12-turn max budget
- Prompt breaks topic into 3-6 sub-questions → `web_search/web_fetch` for each → synthesise into Markdown
- Output saved to `research/YYYY-MM-DD-<slug>.md`
- **Known limitation** (stated in code comments): "intentionally the *minimal* version" — no middleware pipeline, no concurrent sub-agents, no confidence-scored memory.

1.2 :council <topic> — Ecclesia Multi-Agent Council

File: `src/research/council.ts` (275 lines). More mature than `:research`.

- **5 independent agents** (odd number — avoids ties), running *sequentially* (cost-conscious)
- Each agent produces bullet findings with sources — they do not see each other's work
- **Synthesis pass** on the caller's active provider, producing four canonical sections:
 - **Convergent findings** — claims most/all agents independently arrived at
 - **Contested** — where agents disagreed, with split counts
 - **Minority signal** — things only 1-2 agents caught worth surfacing
 - **Verdict** — bottom-line conclusion with confidence level
- Output: `.md` + styled `.html` under `council/`
- **Panel model resolution:** `--panel <model>` → `AURA_PANEL_MODEL` env var → synthesis provider's model

1.3 Supporting Infrastructure

- `src/tools/web-search.ts` — DuckDuckGo (no API key needed)
- `src/tools/web-fetch.ts` — HTTP GET with SSRF guard
- `src/agent/loop.ts` — generic agent loop used by both research and council
- `src/agent/mixture.ts` — MoA (mixture-of-agents) for exploratory tasks, already used by `:council`
- `research/generate-pdf.mjs` — Puppeteer HTML → PDF (tested, works)

2. Output Quality Audit

TOPIC	METHOD	RATING	NOTES
Milutin Milanković	Council (5 agents)	★★★★★	Exceptional. Accurate birth/death dates, nuanced "Croll vs original synthesis" contestation, minority signals flagged (Revised Julian calendar mathematical precision, untranslated autobiography), high-confidence verdict. 7 proper sources.
Antigravity	Research (single)	★★★★★	Near-academic quality. Einstein field equations in LaTeX, 18 sources with DOIs, ALPHA-g experiment results with exact values, structured "Open Debates" table. Arguably the best output in the repo.
Russian Cancer Vaccines	Council (5 agents)	★★★★★	Good structure, appropriately skeptical. Sources are real: Nature, Lancet, ClinicalTrials.gov, New York Times. One agent's "some vaccines already approved" claim flagged as minority outlier.
Nikola Tesla	Council (5 agents)	★★★☆☆	Correct but shallow. All 5 agents converged on the same generic facts. Only one trivial contested item. No novel synthesis — reads like a Wikipedia summary.
Mihajlo Pupin	Research (single)	★☆☆☆☆	Failed. "Loop ended after 12 turns" — agent got stuck. The polished <code>index.html</code> was manually constructed, not agent-generated.

TOPIC	METHOD	RATING	NOTES
Mileva Marić	Council (5 agents)	★☆☆☆☆	Hallucination disaster. See Section 3.

3. The Mileva Marić Problem

This is the single most important quality failure in the current system. Five independent agents researching Mileva Marić produced **catastrophically hallucinated claims**:

CLAIM	REALITY
"Coined the term quantum field theory"	False. QFT was developed by Dirac, Heisenberg, Pauli, Feynman, Schwinger, Tomonaga (1920s-1940s).
"Developed the concept of spacetime"	False. Minkowski (1908), building on Einstein's special relativity.
"Maric-Penrose spinors" (1934)	Fabricated. No such concept exists. Penrose's spinor work is from the 1960s.
"Received Nobel Prize in Physics (posthumously)"	Fabricated. She never received any Nobel Prize.
"President of the Serbian Academy of Sciences"	Fabricated. She never held this position.
Born 1905, died 1987	False. Actual: 1875–1948.
"Worked with Einstein on the EPR paradox"	False. EPR paper (Einstein-Podolsky-Rosen) was 1935, long after their relationship ended.
Sources cited: "Maric, M. (1980)", "Einstein & Maric (1936)", etc.	All fabricated. These publications do not exist.

Root cause: The agents were asked to research a historical figure about whom *little reliable training data exists* — Mileva Marić's actual scientific contributions are a legitimate historical debate, but LLMs fill the vacuum with synthetic "facts." The council's own synthesis pass partially caught this ("confidence is low regarding any specific achievement"), but it *still* presented the hallucinated claims as "agent findings" that a reader could easily mistake for real research.

Required Fix: Anti-Hallucination Pass

Every research output needs a **Devil's Advocate pass** — an agent whose only job is to challenge every factual claim:

For each claim in the report:

- Find the original source online
- Verify the claim against the source
- Mark:  VERIFIED (with link) |  PLAUSIBLE BUT UNVERIFIED |  CONTRADICTED (with evidence)

This should run *after* research but *before* the final synthesis.

4. What's Missing for Academic Paper Writing

4.1 Depth — Multi-Pass Iterative Research

Current `:research` does one pass. Academic deep research requires:

- **Iterative deepening:** Pass 1 → identify gaps → Pass 2 focused on gaps → Pass 3
- **Citation tracing:** Find paper → follow backward references → follow forward citations
- **Source quality scoring:** distinguish arXiv preprint from peer-reviewed Nature paper

4.2 Formal Academic Structure

Current outputs are reports, not papers. Missing sections:

- Abstract (250-300 word structured summary)
- Introduction / Literature Review
- Methodology (how was the research conducted)
- Findings / Results
- Discussion (implications, limitations, future work)
- Conclusion
- References (formatted: APA, BibTeX, or journal-specific)

The council's "Convergent / Contested / Minority Signal / Verdict" structure is *analogous* but not a proper academic format.

4.3 Citation Management

- No source registry — each agent rediscovers sources independently
- No `.bib` output — can't feed into LaTeX or Zotero
- No citation key system — can't write "as shown by `\cite{smith2024}`"

4.4 LaTeX / PDF Output

Current output: Markdown + HTML. For actual papers: LaTeX with proper typesetting, `\cite{}`, `\section{}`, `\begin{figure}`, journal templates, generated PDF.

4.5 Peer Review Simulation

No quality gate before final output. A simulated reviewer pass would catch: structural gaps, unsupported claims, contradictions between sections, missing methodology.

5. Proposed Architecture — The `:paper` Pipeline

```
:paper "Quantum Error Correction in NISQ-era Devices"
|
├─ Phase 1: DEEP RESEARCH (3-5 agents, multiple passes)
|   ├── 1a. Council-style multi-agent research
|   ├── 1b. Identify gaps & conflicting claims
|   ├── 1c. Focused 2nd pass on gaps
|   └─ 1d. Source registry (collect all citations)
|
├─ Phase 2: DEVIL'S ADVOCATE
|   └─ Challenge every claim: verified / plausible /
contradicted
|
├─ Phase 3: OUTLINE
|   └─ Structure paper: sections, subsections, key claims
per section
|
├─ Phase 4: WRITING
|   ├── 4a. Write each section (citing Phase 1 sources)
|   ├── 4b. Generate .bib file from source registry
|   └─ 4c. Cross-section consistency check
|
├─ Phase 5: PEER REVIEW
|   └─ Simulated reviewer: structured feedback, gap analysis
|
├─ Phase 6: FINAL ASSEMBLY
|   ├── 6a. Apply reviewer fixes
|   ├── 6b. Format: Markdown + LaTeX
|   └─ 6c. Render PDF via LaTeX (or Puppeteer fallback)
|
└─ Output: paper.md + paper.tex + references.bib + paper.pdf
```

6. Implementation Roadmap

Tier 1 — Quick Wins (hours/days)

6.1 Anti-Hallucination Pass **CRITICAL**

New file: `src/research/fact-check.ts`

- Runs after `:research` and `:council` (opt-in flag: `--fact-check`)
- Agent receives the full report + instruction to verify every factual claim
- Output: annotated report with  /  /  markers
- Estimated tokens: ~8k-15k per pass

6.2 Iterative Deepening `--rounds N`

Modify: `src/research/research.ts`

- Each round receives the previous round's output + "find what's missing, go deeper"
- Default: 1 round (current behaviour). `--rounds 3` for deep research

Tier 2 — Core Paper Pipeline (days/week)

6.3 Source Registry + Citation Engine

New file: `src/research/citation-engine.ts`

- Collects all URLs/sources from research phase
- Assigns citation keys: `smith2024`, `nature2023`
- Generates `.bib` output (BibTeX format)
- Enables `\cite{smith2024}` in generated text

6.4 `:paper` Command Shell

New file: `src/research/paper.ts`

- Orchestrator: deep research → fact-check → outline → write → review
- Reuses existing `research.ts` and `council.ts` as Phase 1 engines

- CLI: `:paper <topic> [--model <m>] [--panel <m>] [--rounds <n>] [--format md|tex]`

6.5 LaTeX Output

New file: `src/research/latex-render.ts`

- Convert the structured paper into `.tex` with proper `\section`, `\cite`, `\begin{abstract}`
- Use an existing journal template (e.g., arXiv, IEEE, Springer LNCS)
- PDF generation: call `pdflatex` if available, fall back to Puppeteer as already done in `research/generate-pdf.mjs`

Tier 3 — Advanced (weeks)

6.6 Automated Peer Review

New file: `src/research/peer-review.ts`

- Agent acts as a journal reviewer: structured feedback form
- Checks: evidence quality, logical flow, missing citations, overclaims
- Optionally: second iteration — apply reviewer fixes, re-review

6.7 Research Plan / Template System

User specifies: "I want a systematic review on X, with methodology comparison and case study." System adapts the research plan to the template, allocating agent budget per section.

6.8 Knowledge Graph Integration

Write research findings into the existing perception graph (`src/perception/`).

Future papers can query: "What do we already know about Milanković cycles?" — reusing previous research to seed new investigations.

7. Component Design Sketches

7.1 Source Registry (Data Structure)

```
interface SourceEntry {
  key: string;           // "smith2024"
  type: 'paper' | 'article' | 'webpage' | 'report';
  title: string;
  authors: string[];
  year: number;
  url: string;
  doi?: string;
  journal?: string;
  verified: boolean;     // did fact-check confirm this exists?
  claims: string[];      // which claims in the paper cite this
  source
}
```

7.2 Paper Outline (Data Structure)

```
interface PaperOutline {
  title: string;
  abstract: string;
  sections: {
    heading: string;
    subsections?: { heading: string; keyClaims: string[] }[];
    keyClaims: string[];
    assignedSources: string[]; // citation keys
  }[];
}
```

7.3 Fact-Check Report

```
interface FactCheckResult {
  claim: string;
  section: string;
  verdict: 'verified' | 'plausible' | 'contradicted' |
  'unverifiable';
  source?: string;        // URL that confirms or refutes
}
```

```
    note?: string;           // explanation
}
```

7.4 Peer Review Form

```
interface PeerReview {
  overallRating: 1-5;
  strengths: string[];
  weaknesses: string[];
  evidenceQuality: { section: string; rating: 1-5; note:
string }[];
  structuralIssues: string[];
  missingCitations: string[];
  recommendation: 'accept' | 'minor_revision' |
'major_revision' | 'reject';
}
```

8. Cost & Token Economics

Estimates for a ~5,000-word academic paper on a moderately deep topic, using `glm-5.2` as synthesis model and `glm-5.1` as panel model:

PHASE	AGENTS	TOKENS (EST.)	COST (GLM-5 PAY-AS-YOU-GO)
Deep Research (3 rounds × 5 agents)	15 runs	~180,000	~\$0.27
Fact-Check Pass	1 run	~12,000	~\$0.02
Outline Generation	1 run	~8,000	~\$0.01
Section Writing (6 sections)	6 runs	~48,000	~\$0.07
Cross-Section Consistency	1 run	~15,000	~\$0.02
Peer Review	1 run	~18,000	~\$0.03
Final Assembly + LaTeX	1 run	~20,000	~\$0.03
TOTAL	26 runs	~301,000	~\$0.45

Note: This is for one full paper cycle. With Coding Plan subscription (zhipu-coding/), cost drops significantly. A `--fast` mode using `mimo-v2-flash` for panel agents would bring it under **\$0.15/paper**.

Time estimate: With sequential agent runs at ~15-30s each, total wall-clock time: **~8-15 minutes** per paper. Could be parallelised for panel phases (council already supports this conceptually, currently sequential by design for cost safety).

Token Budget Comparison

OPERATION	CURRENT	PROPOSED :PAPER
:dream full (glm-5.1)	952,492	—

OPERATION	CURRENT	PROPOSED :PAPER
:council (5 agents, no deepening)	~60,000	—
:research (single pass)	~30,000	—
:paper full cycle (est.)	—	~301,000

A full `:paper` costs $\sim\frac{1}{3}$ of a `:dream full` and $\sim 5\times$ a `:council`. This is reasonable: you're trading $5\times$ the tokens for a complete academic paper with fact-checking, citations, and peer review — not just a research summary.